

Enrico Antonini

Senior Energy System Modeller · Team and Project Manager, Open Energy Transition · Verona, Italy
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Summary

I work on clean technologies, electricity grids and energy systems for the energy transition, including how they interact with weather and climate. I use engineering, modelling, optimisation and data to turn complex problems into practical solutions and clear, rigorous analysis. My experience spans more than ten years across research and applied energy planning, including renewable technologies, electricity-system reliability, energy-system optimisation and open-source software. At Open Energy Transition, I lead a modelling team and coordinate technical projects, combining hands-on technical development and analysis with project management and interdisciplinary collaboration.

Experience

- Jan 2025 – present **Open Energy Transition**
Verona, Italy (remote)
Project Manager · Jan 2026 – present
- Coordinate DemandCast, the Open Energy Benchmark and a PyPSA-Eur study of clean firm technologies in Europe.
 - Coordinate Open Energy Transition's activities across the Horizon Europe project AVANGUARD and with project partners.
- Team Manager** · Apr 2025 – present
- Lead a team of five focused on integrated planning for renewable resources, turning model results into actionable insights across the organisation.
- Senior Energy System Modeller** · Jan 2025 – present
- Lead the development of DemandCast, and help develop the Open Energy Benchmark and the PyPSA-Eur study of clean firm technologies in Europe.
- Jan 2023 – Dec 2024 **CMCC Foundation – Euro-Mediterranean Center on Climate Change**
Milan, Italy
Junior Scientist
- Created a modelling framework that estimates weather- and climate-driven power supply and demand time series for energy system analyses.
 - Studied the impacts of climate change on renewable generation, and the reliability of deeply decarbonised energy systems under weather and climate stressors.
- Mar 2019 – Dec 2022 **Carnegie Institution for Science**
Stanford, CA, USA
Postdoctoral Research Scientist
- Studied the physics of wind power extraction by regional-scale wind farms: how it changes with farm size, and the scales at which that change occurs.
 - Investigated how to resolve the integration challenges of a large-scale expansion of wind and solar in decarbonised electricity systems.

Oct 2018 – Jan 2019	University of Toronto Toronto, Canada Postdoctoral Fellow Applied the CFD-based layout optimisation method from my PhD, maximising for the first time the annual energy production of a wind farm in complex terrain by changing turbine positions.
Oct 2018 – Jan 2019	Sheridan College Toronto, Canada (part-time) Research Assistant Studied the performance of an innovative shrouded vertical-axis wind turbine with CFD models, and assessed design changes for the prototype later built and tested.

Education

Sept 2014 – Sept 2018	PhD, Mechanical and Industrial Engineering University of Toronto · Toronto, Canada Thesis: <i>CFD-based methodology for wind farm layout optimization</i> . Supervisors: Prof. Cristina H. Amon, Dr. David A. Romero.
Oct 2010 – Mar 2013	MSc, Mechanical Engineering University of Padua · Padua, Italy · 110/110 with honours Thesis: <i>Development of a prescribed expanding vortex wake model for HAWTs</i> . Supervisors: Prof. Ernesto Benini, Prof. Jens Nørkær Sørensen, Dr. Marco Raciti Castelli.
Oct 2007 – Sept 2010	BSc, Mechanical Engineering University of Padua · Padua, Italy · 110/110 with honours Thesis: <i>Optimized management of a power-split transmission for agricultural tractors</i> . Supervisors: Prof. Alarico Macor, Dr. Antonio Rossetti.

Publications

Journal articles

14. **E.G.A. Antonini**, A. Di Bella, I. Savelli, L. Drouet, M. Tavoni, "Weather- and climate-driven power supply and demand time series for power and energy system analyses", *Scientific Data*, 11, 1324, 2024.
13. T.H. Ruggles, E. Virguez, N. Reich, J. Dowling, H. Bloomfield, **E.G.A. Antonini**, S.J. Davis, N.S. Lewis, K. Caldeira, "Planning reliable wind- and solar-based electricity systems", *Advances in Applied Energy*, 15, 100185, 2024.
12. **E.G.A. Antonini**, E. Virguez, S. Ashfaq, L. Duan, T.H. Ruggles, K. Caldeira, "Identification of reliable locations for wind power generation through a global analysis of wind droughts", *Communications Earth & Environment*, 5 (1), 103, 2024.
11. D.A. Romero, S. Hasanpoor, **E.G.A. Antonini**, C.H. Amon, "Predicting wind farm wake losses with deep convolutional hierarchical encoder-decoder neural networks", *APL Machine Learning*, 2 (1), 016111, 2024.
10. **E.G.A. Antonini**, T. Ruggles, D.J. Farnham, K. Caldeira, "The quantity-quality transition in the value of expanding wind and solar power generation", *iScience*, 25 (4), 104140, 2022.
9. **E.G.A. Antonini**, K. Caldeira, "Spatial constraints in large-scale expansion of wind power plants", *Proceedings of the National Academy of Sciences*, 118 (27), e2103875118, 2021.
8. A. Dhoot, **E.G.A. Antonini**, D.A. Romero, C.H. Amon, "Optimizing wind farms layouts for maximum energy production using probabilistic inference: Benchmarking reveals superior computational efficiency and scalability", *Energy*, 223, 120035, 2021.
7. **E.G.A. Antonini**, K. Caldeira, "Atmospheric pressure gradients and Coriolis forces provide geophysical limits to power density of large wind farms", *Applied Energy*, 281, 116048, 2021.
6. **E.G.A. Antonini**, D.A. Romero, C.H. Amon, "Optimal design of wind farms in complex terrains using computational fluid dynamics and adjoint methods", *Applied Energy*, 261, 114426, 2020.
5. **E.G.A. Antonini**, D.A. Romero, C.H. Amon, "Improving CFD wind farm simulations incorporating wind direction uncertainty", *Renewable Energy*, 133, 1011–1023, 2019.
4. **E.G.A. Antonini**, D.A. Romero, C.H. Amon, "Continuous adjoint formulation for wind farm layout optimization: A 2D implementation", *Applied Energy*, 228, 2333–2345, 2018.

3. **E.G.A. Antonini**, D.A. Romero, C.H. Amon, "Analysis and modifications of turbulence models for wind turbine wake simulations in atmospheric boundary layers", *Journal of Solar Energy Engineering*, 140 (3), 031007, 2018.
2. **E.G.A. Antonini**, G. Bedon, S. De Betta, L. Michelini, M. Raciti Castelli, E. Benini, "Innovative discrete-vortex model for dynamic stall simulations", *AIAA Journal*, 53 (2), 479–485, 2015.
1. G. Bedon, **E.G.A. Antonini**, S. De Betta, M. Raciti Castelli, E. Benini, "Evaluation of the different aerodynamic databases for vertical axis wind turbine simulations", *Renewable and Sustainable Energy Reviews*, 40, 386–399, 2014.

Refereed conference papers

3. K. Steijn, V.P. Goli, **E.G.A. Antonini**, "DemandCast: Global hourly electricity demand forecasting", *NeurIPS 2025 Workshop: Tackling Climate Change with Machine Learning*, 2025.
2. **E.G.A. Antonini**, T. Ruggles, D.J. Farnham, K. Caldeira, "Meeting electricity demand with distributed wind and solar generation: System flexibility drives optimal siting", *Proceedings of the ASME International Mechanical Engineering Congress and Exposition*, IMECE2021-70678, 2021.
1. **E.G.A. Antonini**, D.A. Romero, C.H. Amon, "Analysis and modifications of turbulence models for wind turbine wake simulations in atmospheric boundary layers", *Proceedings of the ASME International Mechanical Engineering Congress and Exposition*, IMECE2016-67353, 2016.

Reports

1. **E.G.A. Antonini**, S. Krishna, D. Lerede, "The State of Solvers for Energy Planning – Results of the 2025 Open Energy Benchmark", *Open Energy Transition*, 2026.

Presentations

Oral presentations

14. J. Hall, **E.G.A. Antonini**, D. Paoletta, M. Miltenberger, H. van der Weijde (moderator), "Benchmarking optimization solvers for energy system models: 2025 results", *Open Energy Transition webinar*, Online, 2026.
13. K. Steijn, V.P. Goli, **E.G.A. Antonini**, "DemandCast: Global hourly electricity demand forecasting", *NeurIPS 2025 Workshop: Tackling Climate Change with Machine Learning*, San Diego, CA, USA, 2025.
12. **E.G.A. Antonini**, A. Di Bella, L. Drouet, M. Tavoni, "The role of hydropower in renewable-rich energy systems under climate change", *International Energy Workshop*, Bonn, Germany, 2024.
11. **E.G.A. Antonini**, A. Di Bella, L. Drouet, M. Tavoni, "More than a century of weather- and climate-dependent power supply and demand time series", *Openmod Workshop*, Grenoble, France, 2024.
10. **E.G.A. Antonini**, E. Virgüez, S. Ashfaq, L. Duan, T.H. Ruggles, K. Caldeira, "Historical analysis of global distribution of and trends in wind droughts", *EGU General Assembly*, Vienna, Austria, 2023.
9. **E.G.A. Antonini**, K. Caldeira, "Geophysical constraints to large wind farm development", *NAWEA/WindTech Conference*, University of Delaware, Newark, DE, USA, 2022.
8. **E.G.A. Antonini**, T. Ruggles, D.J. Farnham, K. Caldeira, "The quantity-quality transition in the value of expanding wind and solar power generation", *Macro Energy Systems Workshop*, Stanford University, Stanford, CA, USA, 2022.
7. **E.G.A. Antonini**, T. Ruggles, D.J. Farnham, K. Caldeira, "Meeting US electricity demand with distributed wind and solar generation: System flexibility drives optimal siting", *ASME International Mechanical Engineering Congress and Exposition*, Virtual conference, 2021.
6. **E.G.A. Antonini**, K. Caldeira, "How atmospheric pressure gradients and Coriolis forces control the power density of large wind farms", *Wind Energy Science Conference*, Hannover, Germany, 2021.
5. **E.G.A. Antonini**, D.A. Romero, C.H. Amon, "Computational-fluid-dynamics-based methodology for wind farm layout optimization", *Seminar Series, Carnegie Institution for Science*, Stanford, CA, USA, 2018.
4. **E.G.A. Antonini**, D.A. Romero, C.H. Amon, "Continuous adjoint formulation for wind farm layout optimization", *8th MIE Symposium*, University of Toronto, Toronto, ON, Canada, 2017.
3. **E.G.A. Antonini**, D.A. Romero, C.H. Amon, "Analysis and modifications of turbulence models for wind turbine wake simulations in atmospheric boundary layers", *ASME International Mechanical Engineering Congress and Exposition*, Phoenix, AZ, USA, 2016.
2. **E.G.A. Antonini**, D.A. Romero, C.H. Amon, "Enhancement of CFD wind farm simulations through introduction of wind direction uncertainty", *7th MIE Symposium*, University of Toronto, Toronto, ON, Canada, 2016.
1. **E.G.A. Antonini**, D.A. Romero, C.H. Amon, "Implementation and simulation of wind turbines with the OpenFOAM solver using the actuator disk approach", *6th MIE Symposium*, University of Toronto, Toronto, ON, Canada, 2015.

Posters

13. **E.G.A. Antonini**, E. Virgüez, S. Ashfaq, L. Duan, T.H. Ruggles, K. Caldeira, "Historical analysis of global distribution of and trends in wind droughts", *International Conference on Energy and Meteorology*, Padua, Italy, 2023.
 12. K. Caldeira, A. Li, E. Virgüez, **E.G.A. Antonini**, J.A. Dowling, L. Duan, M.O. Dioha, N. Reich, N.S. Lewis, S.J. Davis, T. Ruggles, S. Ashfaq, "A macro energy modeling framework for transparent analysis of implications of energy system assumptions", *AGU Fall Meeting*, Chicago, IL, USA, 2022.
 11. **E.G.A. Antonini**, E. Virgüez, S. Ashfaq, L. Duan, K. Caldeira, "Characterizing geophysical limits to wind power reliability", *AGU Fall Meeting*, Chicago, IL, USA, 2022.
 10. **E.G.A. Antonini**, T. Ruggles, D.J. Farnham, K. Caldeira, "The quantity-quality transition in the value of expanding wind and solar power generation", *Macro Energy Systems Workshop*, Stanford University, Stanford, CA, USA, 2022.
 9. **E.G.A. Antonini**, K. Caldeira, "Replenishing the wind: Atmospheric physics explains limits to energy extraction and spatial constraints in large-scale expansion of wind power plants", *AGU Fall Meeting*, New Orleans, LA, USA, 2021.
 8. **E.G.A. Antonini**, T. Ruggles, D.J. Farnham, K. Caldeira, "Strategic site selection of wind and solar power plants in deep decarbonization scenarios for electricity systems", *AGU Fall Meeting*, New Orleans, LA, USA, 2021.
 7. **E.G.A. Antonini**, K. Caldeira, "How atmospheric pressure gradients and Coriolis forces control the power density of large wind farms", *AGU Fall Meeting*, San Francisco, CA, USA, 2020.
 6. M. Hauser, T. Ruggles, C. Henry, K. Caldeira, R. Peer, **E.G.A. Antonini**, "Cost sensitivity of electricity systems to the shape of electricity demand curve: A Sub-Saharan Africa example", *AGU Fall Meeting*, San Francisco, CA, USA, 2020.
 5. T. Ruggles, D.J. Farnham, C. Henry, R. Peer, L. Duan, **E.G.A. Antonini**, M. Hauser, N. Lewis, J.A. Dowling, K. Rinaldi, S.J. Davis, D. Tong, K. Caldeira, "Electrofuels and curtailment of wind and solar power", *AGU Fall Meeting*, San Francisco, CA, USA, 2020.
 4. **E.G.A. Antonini**, K. Caldeira, "Limits of electricity generation from wind: Characterizing transitional scales in wind farm power density", *AGU Fall Meeting*, San Francisco, CA, USA, 2019.
 3. O. Tayyara, **E.G.A. Antonini**, D.A. Romero, C.H. Amon, "CFD modeling of after-market rotor attachments performance on horizontal axis wind turbines", *9th MIE Symposium*, University of Toronto, Toronto, ON, Canada, 2018.
 2. **E.G.A. Antonini**, D.A. Romero, C.H. Amon, "Continuous adjoint formulation for wind farm layout optimization", *9th MIE Symposium*, University of Toronto, Toronto, ON, Canada, 2018.
 1. D. Guirguis, S.Y.D. Yamani, **E.G.A. Antonini**, J.Y.J. Kuo, D.A. Romero, C.H. Amon, "Wake modelling and design optimization of wind farms", *Institute of Sustainable Energy Research Symposium*, University of Toronto, Toronto, ON, Canada, 2016.
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Teaching

Wind Power — Guest lecturer

Graduate course, University of Toronto · Fall 2018

Taught the fluid mechanics of wind farms, with emphasis on wakes and their interaction with the atmospheric boundary layer, using real wind farms as worked examples.

Summer STARS — Guest speaker

Carnegie Academy for Science Education · July 2021

Spoke to high-school interns about my research at Carnegie and about careers in science, technology, engineering and mathematics.

Wind Power — Teaching assistant

Graduate course, University of Toronto · Fall 2017 and 2018

Redesigned the three problem sets and their marking schemes, and gave students detailed feedback.

Thermal Energy Conversion — Teaching assistant

Undergraduate course, University of Toronto · Winter 2018

Ran a laboratory on a two-stage intercooled air compressor driven by a steam engine, and marked lab reports and assignments.

Alternative Energy Systems — Teaching assistant

Graduate course, University of Toronto · Fall 2016 and 2017

Designed marking schemes for problem sets and simulation reports, and presented the solutions.

Fluid Mechanics I — Teaching assistant

Undergraduate course, University of Toronto · Fall 2016

Ran a laboratory on head losses in pipes and fittings, and marked lab reports and exams.

Mentoring

Alice Di Bella — PhD, Polytechnic University of Milan, 2023–2025

Project: *Least-cost carbon-neutral scenarios for the European energy system*

Omri Tayyara — MEng, University of Toronto, 2017–2018

Project: *CFD modelling of after-market rotor attachments on wind turbines*. First position after the degree: PhD student, University of Toronto

Danyal Rehman — BAsC, University of Toronto, 2017

Project: *Wind farm power optimisation using adaptive yaw control*. First position after the degree: Graduate student, MIT

Harmit Komal — MEng, University of Toronto, 2015–2016

Project: *Modelling wind turbine wakes in complex terrain*. First position after the degree: Project Engineer, Environment and Climate Change Canada

Adithya Dhoot — MAsC, University of Toronto, 2015–2016

Project: *Wind farm layout optimisation using probabilistic inference*. First position after the degree: Software Engineer, Autodesk

Service & recognition

Guest editor, Proceedings of the National Academy of Sciences (2021)

Primary convener, AGU Fall Meeting session “Net-Zero Emissions Energy Systems: Geophysical Constraints, Consequences, and Opportunities” (2022)

Judge, AGU Fall Meeting student presentation awards (2019, 2021)

Reviewer for Joule, Applied Energy, Energy, Renewable Energy, Communications Earth & Environment, Energy Conversion and Management, Wind Energy and other journals (since 2016)

Diversity, equity and inclusion advisory team, Carnegie Institution for Science (2021–2022)

Faculty search committee, early-career representative, Carnegie Institution for Science (2020–2021)

Funding

Gates Ventures postdoctoral funding, US\$253,380 (2019–2023)

University of Toronto MIE graduate scholarship, CA\$139,843 (2014–2018)

Hatch graduate scholarship for sustainable energy research, CA\$20,000 (2016–2018)

Metcalf family fellowship for sustainable energy research, CA\$6,000 (2017–2018)

In the press

“Climate change, il rischio che cambino i venti”, *ANRA*, 10 Jul 2024.

“Discovering wind droughts and their impacts on energy supply”, *Foresight*, 12 Jun 2024.

“Where the wind blows for strong renewable energy investment”, *Nature Italy*, 29 Apr 2024.

“Study identifies where wind is most reliable for generating power”, *The Guardian*, 18 Apr 2024.

“Protecting self-driving cars from cosmic rays, size limits for wind farms”, *Physics World podcast*, 29 Jul 2021.

“Research finds optimal size for windfarms”, *The Guardian*, 27 Jul 2021.

“L'uomo del vento: “Così si ottimizza l'eolico””, *la Repubblica*, 9 Jul 2021.

“Optimal size for wind farms is revealed by computational study”, *Physics World*, 8 Jul 2021.

"Come migliorare il rendimento dei grandi campi eolici del futuro", *QualEnergia*, 5 Jul 2021.

"How to build a better wind farm", *ScienceDaily*, 28 Jun 2021.

"Enrico, il ricercatore green in fuga che sogna di ritornare", *L'Arena*, 2 Jan 2021.

Skills

- **Energy system modelling:** PyPSA, PyPSA-Eur, linopy, atlite; optimisation solvers (HiGHS, Gurobi, CBC, GLPK, SCIP); Snakemake workflows
 - **Data and machine learning:** Python (pandas, xarray, scikit-learn, XGBoost), Jupyter; weather and climate data (ERA5, CMIP6)
 - **Software engineering:** Git and GitHub, continuous integration with GitHub Actions, pixi and conda, pytest, ruff, pre-commit, Docker
 - **Management:** project coordination, including Horizon Europe projects; team leadership
 - **Earlier research tools:** OpenFOAM, Ansys Fluent and CFX, WRF, Fortran, C++, MATLAB
 - **Languages:** Italian (native), English (fluent)
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Professional memberships

- American Society of Mechanical Engineers (ASME)
 - American Geophysical Union (AGU)
 - European Geosciences Union (EGU)
 - Macro Energy Systems (MES) community
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Training

- Bystander intervention — Carnegie Institution for Science, 2021
- How to conduct an inclusive search in STEM — Carnegie Institution for Science, 2020
- Laboratory training: performance of a two-stage air compressor — University of Toronto, 2018
- Laboratory training: head losses in pipe systems — University of Toronto, 2016
- Ethics in research — University of Toronto, 2015